Incident Report: Password Spraying Detection

Alert Title

Multiple Failed Logins Across Different Accounts from Single IP

Summary

Multiple failed login attempts were observed targeting several user accounts from a single external IP address within a short time window. Each account experienced a low number of attempts, and no successful authentication was recorded. This pattern is consistent with a password spraying attack, where an attacker attempts common passwords across multiple accounts to avoid lockouts.

Initial Findings

- Source IP: `185.76.23.44`
- Time Window: 09:10 PM – 09:18 PM
- Total Failed Attempts: 18
- Accounts Targeted:
  - ali.raza
  - sara.khan
  - usman.shahid
  - fatima.noor
  - ahmed.ali
- Successful Login: No
- Location: Netherlands

Key Observations

- All login attempts originated from a single external IP
- Multiple distinct user accounts were targeted
- Low number of attempts per account

- Activity occurred within a short time frame (8 minutes)
- Source IP belongs to a hosting provider, increasing suspicion
- Pattern aligns with password spraying behavior

Investigation

Failed Login Analysis

```
SecurityEvent
| where EventID == 4625
| summarize FailedCount = count(), UniqueAccounts = dcount(Account)
    by IPAddress, bin(TimeGenerated, 10m)
| where FailedCount >= 15 and UniqueAccounts >= 5
| sort by FailedCount desc
```

Result: Confirmed multiple failed login attempts across different accounts from the same IP within a short time window.

Evidence

- 18 failed login events (Event ID 4625)
- 5 distinct user accounts targeted
- Single source IP involved
- No successful login observed
- Unusual geographic location

Conclusion

This incident is classified as a **True Positive (Password Spraying Attempt)**.
Although no successful authentication occurred, the attack pattern indicates an active attempt to gain unauthorized access to multiple user accounts.

Severity

Medium

Response Actions

- Blocked the malicious source IP
- Enabled enhanced monitoring for targeted user accounts
- Recommended password reset for affected users
- Reviewed authentication logs for any delayed or missed successful attempts
- Escalated findings for further monitoring and detection tuning

Analyst Note

This activity highlights the importance of implementing strong password policies and multi-factor authentication (MFA). Detection rules can be further tuned to identify similar patterns early and prevent potential account compromise.